DEFORMATION THEORY

github.com/danimalabares/stack

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1. Introduction

[**?**] has a great way of motivating deformation theory via the moduli problem. Here let's just put the following important meaning for "deformation theory" (as opposed to studying flat families!):

> "*Deformation theory* is the study of infinitesimal deformaitons as a tool to understand the local structure of the moduli space. The goal is to be able to describe the restriction of the universal family to a small neighborhood of $m \in \mathcal{M}$, or, more precisely, its restriction to the germ of M at m.
>
> What is interesting here is that we can study order and infinitesimal deformations even though the functor F is not representable or simply we don't yet know it is. This is the most frequent case. Such an investigation will reveal the infinitesimal properties at $[X]$ of a yet unknown global structure on M which will be hopefully understood at a subsequent stage of the investigation. In order words it turns out to be possible and convenient to separate the *global moduli problem* from the *local moduli problem*, and deformation theory studies the latter, with the purpose of constructing a family of deformations of a given object parametrized by the spectrum of a local ring, and having properties as close as possible to a universal property."

2. Extensions

In this section I discuss extensions of rings and their geometric counterpart.

Definition 2.1. Let $A \to R$ be a ring homomorphism (i.e. R is an A-algebra). An *A-extension of R by I* is a short exact sequence

$$0 \longrightarrow I \longrightarrow E \xrightarrow{\varphi} R \longrightarrow 0.$$

where E is an R'-algebra and φ is a morphism of A-algebras whose kernel is I. We also require that $I^2 = (0)$; this condition implies that I has the sturcture of R-module.

Example 2.2.

$$0 \longrightarrow (\varepsilon) \longrightarrow A[\varepsilon] \longrightarrow A \longrightarrow 0$$

where $A[\varepsilon]$ is the algebra of dual numbers over A.

Without explaining details, let me for now just say that there are notions of morphism of extensions, pushforward, pullbacks, and even a sum with makes the category of extensions of A-algebras R by I, denoted by $\mathrm{Ex}_A(R, I)$, an R-module. More precisely:

Proposition 2.3. [**?**, Proposition 1.1.4].

Next in line: define the first cotangent module (so that I may understand the grading of the first cotangent module of a SR ring. . .)

3. Deformations of schemes

The following definition is from [lucas-defos], which in turn comes from [**?**]

Definition 3.1. Let X be an algebraic $\mathbb{C}$-scheme.

(1) A *deformation* of X is a Cartesian diagram ξ

$$\begin{tikzcd} X \arrow[r] \arrow[d] & \mathcal{X} \arrow[d, "\pi"] \\ \operatorname{Spec}\mathbb{C} \arrow[r, "s"] & S \end{tikzcd}$$

where π is a flat surjective morphism of algebraic $\mathbb{C}$-schemes and S is connected. (Recall that flatness accounts for "continuity".)

(2) A *local deformation* of X is a deformation ξ where $S = \operatorname{Spec} A$ for A a noetherian local $\mathbb{C}$ algebra with residue field $\mathbb{C}$.

(3) An *infinitesimal deformation of* X is a local deformation with A an artinian local $\mathbb{C}$-algebra with residue field $\mathbb{C}$. X is called *rigid* if all infinitesimal deformations are trivial.

(4) An *inifinitesimal deformation of order* n is an infinitesimal deformation when $S = \operatorname{Spec}(\mathbb{C}[t]/(t^{n+1})$.

4. Kodaira-Spencer map

5. Kodaira-Spencer correspondence

After a few months of the previous text in this section I think I have understood the construction of Kodaira-Spencer map for embedded deformations, which is [**?**, Proposition 3.2.1(ii)].

But I will actually briefly outline a particular case of this, which is easier: [**?**, Examples 3.2.4(ii)]. This the case when $X \subset Y$ is a Cartier divisor. Then X is cut out locally by a set of functions f_i defined on an affine cover U_i of Y. Then we have a line bundle defined by the transition functions $f_{ij} = f_i/f_j$. This is the line bundle $\mathcal{O}_Y(X)$, which by the adjunction formula is also the normal bundle $N_{X/Y}$ of X in Y.

Now consider a first order deformation

$$\begin{tikzcd} X \arrow[r] \arrow[d] & \mathcal{X} \subset Y \times B_\varepsilon \arrow[d, "\pi"] \\ 0 \arrow[r] & B_\varepsilon \end{tikzcd}$$

The great thing about first order deformations is that the structure sheaf of $\mathcal{X}$ on an affine open U_i of Y looks like the tensor product $\mathcal{O}_{U_i} \otimes \mathbb{C}[\varepsilon]/(\varepsilon^2)$ (01I4). By definition of tensor product this just says that sections look like $f + \varepsilon g$ for $f, g \in \Gamma(\mathcal{O}_{U_i})$.

If $\mathcal{X}$ is a Cartier divisor on $Y \times B_\varepsilon$ (it is; but I'm missing a formal proof; the intuition is that an "infinitesimal thickening" $\mathcal{X}$ of X has the same underlying topological space but different scheme structure; 01JL 01I4), it is given as the vanishing locus of some local sections $f_i + \varepsilon g_i$. (Perhaps the f_i coincide with the distinguished section of X.) The line bundle associated to $\mathcal{X}$ is given by the cocycle

$$F_{ij} = f_{ij} + \varepsilon g_{ij}$$

for $g_{ij} = \frac{g_i - f_{ij} g_j}{f_j}$. Indeed, note that $\frac{1}{f_j + \varepsilon g_j} = \frac{1}{f_j} - \varepsilon \frac{g_j}{f_j^2}$ and compute $F_{ij} = (f_i + \varepsilon g_j)/(f_j + \varepsilon g_j)$.

Then

$$f_i + \varepsilon g_i = (f_{ij} + \varepsilon g_{ij})(f_j + \varepsilon g_j)$$

gives

$$g_i = f_{ij} g_j + g_{ij} f_j$$

and since f_j vanishes along X, we are left with $g_i = f_{ij} g_j$ which is how a section of the sheaf $N_{X/Y}$ is constructed by definition of sheaf.

Example 5.1. Fix $g \geq 2$. The dimension of the deformation space of a nonsingular projective curve X is $3g - 3$. This is "the dimension of the moduli space of curves of genus g".

We can compute this number by Riemann-Roch formula on the bundle $-K_X$. Indeed, since X is a curve, $\Omega^1_X = K_X$ and thus $-K_X = T_X$. We get

$$\begin{aligned} h^0(-K) - h^0(K - (-K)) &= \deg(-K) - g + 1 \\ &= -2g + 2 - g + 1 \qquad \substack{\text{degree additive and} \\ \deg K = 2g-2} \\ &= -3g + 3 \end{aligned}$$

Now by Serre duality $h^0(K - (-K)) = h^1(-K) = h^1(T_X)$, and it turns out that a Riemann surface of genus $g \geq 2$ has no holomorphic vector fields, so that $h^0(-K) = h^0(T_X) = 0$.

Exercise 5.2. Let C be a smooth genus g curve which can be embedded in a K3 surface M, and X the family of all deformations of C in M.

(1) Prove that $\dim X \leq g$.
(2) Let $\mathcal{X}_g$ be the space of all curves of genus g (smooth?) which can be possibly embedded to a K3 surface. Prove that each irreducible component Z of $\mathcal{X}_g$ satisfies $\dim_{\mathbb{C}} \leq g + 19$. Deduce that there exists a compact complex curve which cannot be embedded in a K3 surface.

Exercise 5.3. Let C be a smooth curve embedded in a K3 surface X. Show that the dimension of the deformation space of C is $\leq g$.

Proof. The deformation space of a variety is the space of isomorphism classes of deformations as explained above. It turns out that there is a way to associate 1-cocyles of the tangent sheaf to deformations, so that in fact the deformation space Def_1 is isomorphic to $H^1(X, \mathcal{T}_X)$ for any variety X.

For our curve C we thus know that the dimension of the space of deformations (deformations not necessarily contained in X) is $h^1(\mathcal{T}_C)$. The family of deformations of C that are contained in X is the Hilbert scheme of curves with fixed Hilbert polynomial $P(t)$ after quotienting by $\mathbb{C}s$, where s is the section whose vanishing locus is C. This says that the number we are looking for is $h^0(X, \mathcal{O}(C)) - 1$.

Before showing that in fact $h^0(X, \mathcal{O}(C)) = 1 + g$, let's explain why this Hilbert scheme construction actually works. Intuitively, any curve lying in the same component as C would be deformation-equivalent to C — at least we know that it's Hilbert polynomial is the same. But it's not clear what's the deformation involving these two (this question should be addressed!). Further, this doesn't explain Vladimiro's suggesion of "taking quotient by $\mathbb{C}s$". To me it looks like I would just compute the dimension of the tangent space of $\mathrm{Hilb}^P(M)$ at C.

Now I will show that $h^0(X, \mathcal{O}(C)) = 1 + g$ (see [**?**, Lemma 1.2.1, Remark 1.2.2]).

Consider the ideal sheaf exact sequence twisted by $\mathcal{O}(C)$:

$$0 \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}(C) \longrightarrow \mathcal{O}_X(C) \otimes \mathcal{O}_C = \mathcal{O}(C)|_C \longrightarrow 0$$

By X being a K3 we know that $H^1(\mathcal{O}_X) = 0$, so that we have the short exact sequence in cohomology

$$0 \longrightarrow H^0(\mathcal{O}_X) \longrightarrow H^0(\mathcal{O}(C)) \longrightarrow H^0(\mathcal{O}(C)|_C) \longrightarrow 0$$

so that $h^0(\mathcal{O}(C)) = 1 + h^1(\mathcal{O}(C)|_C)$. A version of adjunction formula says $\omega_C \cong (K_X \otimes \mathcal{O}(C)|_C$, and using that $K_X = \mathcal{O}_X$ we obtain $h^0(\mathcal{O}(C)|_C) = h^0(\omega_C) = g$.

To

For the record I put other thoughts I went through in solving this exercise.

Recall from [**?**, p. 146] that the normal bundle $\mathcal{N}$ of a hypersurface of a smooth variety X satisfies $\mathcal{N}^\vee \cong \mathcal{O}_X(-C)|_C$. Taking duals we get that $\mathcal{N} \cong \mathcal{O}_X(C)|_C$.

By adjunction formula $2g-2 = (\mathcal{O}(C), \mathcal{O}(C))$. Applying Riemann-Roch to $\mathcal{O}(C)$ (which is by definition the dual of the ideal sheaf of C, which is a line bundle on X), we obtain that $\chi(\mathcal{O}(C)) = 2 + \frac{1}{2}(\mathcal{O}(C), \mathcal{O}(C))$. Then $\chi(\mathcal{O}(C)) = g + 1$.

Now recall that $\chi(\mathcal{O}(C)) = h^0(\mathcal{O}(C) - h^1(\mathcal{O}(C)) + h^2(\mathcal{O}(C))$. By Serre duality and X being a K3 surface we see that $h^2(\mathcal{O}(C)) \cong h^0(\mathcal{O}(-C))$, which is the ideal sheaf of C. Any section of such a sheaf would vanish along C, and since X is compact we conclude there cannot be any such section.

Now we show that also $h^1(\mathcal{O}(C)) = 0$ to conclude that $h^0(\mathcal{O}(C)) = g + 1$.

We thus conclude that $h^0(\mathcal{O}(C))$

Since C is smooth we can use the normal exact sequence

$$0 \longrightarrow \mathcal{T}_C \longrightarrow \mathcal{T}_X|_C \longrightarrow \mathcal{N} \longrightarrow 0$$

Taking Euler characteristic we see that $\chi(\mathcal{T}_C) + \chi(\mathcal{N}) = \chi(\mathcal{T}_X|_C)$.

This means that we should be done once we compute $\chi(\mathcal{T}_X|_C)$. For this we can use Riemman-Roch formula for coherent sheaves on a curve, which tells us that

$$\deg(\mathcal{T}_X|_C) = \chi(\mathcal{T}_X|_C) - \mathrm{rk}(\mathcal{T}_X|_C) \cdot \chi(\mathcal{O}_C)$$

But of course we know that since C is a curve, by Serre duality we get that $\chi(\mathcal{O}_C) = 1 - g$. (Indeed: $h^1(\mathcal{O}_C) = h^0(\Omega^1_C) := p_a(C)$.) And now the question is what is the degree. Apparently this is just the first Chern class c_1 of the restricted tangent bundle. So what is it? And what is $h^0(\mathcal{T}_C)$ if the genus is 0 or 1, and that's it.

I know that $h^0(\mathcal{T}_X) = 0$ and $h^1(\mathcal{T}_X) = 20$ by X being a K3 surface, but I'm not sure what happens when we restrict to C.

If $g \geq 2$ we know that $h^0(\mathcal{T}_C) = 0$, so that $\boxed{\chi(\mathcal{T}_C) = h^1(\mathcal{T}_C)}$. To compute the latter Euler characteristic, which is given by definition by $\chi(\mathcal{T}_X|_C) = h^0(\mathcal{T}_X|_C) - h^1(\mathcal{T}_X|_C)$, we first note that $h^0(\mathcal{T}_X|_C) = 0$, because this is the dimension of the global holomorphic vector fields on X restricted to C, which is constant along C since X is smooth. And then this number is actually zero by Hodge numbers of a K3 surface.

and taking cohomology long exact sequence we obtain

$$\cdots \longrightarrow H^0(T_X) \longrightarrow H^0(\mathcal{N}) \longrightarrow H^1(T_C) \longrightarrow$$

$$H^1(T_X) \longrightarrow H^1(\mathcal{N}) \longrightarrow \cdots$$

Or is it $h^1(\mathcal{N})$? See [**?**]. If this was the case, then we can use adjunction formula as above to get that $2g - 2 = (\mathcal{N}, \mathcal{N}) = \deg_C \mathcal{N}$. Then we may find $h^0(\mathcal{N})$ via Riemann-Roch:

$$h^0(\mathcal{N}) - h^0(K_C - \mathcal{N}) = \deg \mathcal{N} - g + 1$$

Note that $h^0(N_C - \mathcal{N}) = h^1(-K_C + N + K_C) = h^1(\mathcal{N})$ via Serre duality, and by Riemann-Roch on a surface as above we see that

$$g + 1 = \chi(\mathcal{N}) = h^0(\mathcal{N}) - h^1(\mathcal{N}) \implies h^1(\mathcal{N}) = -g - 1 + h^0(\mathcal{N})$$

so that

$$h^0(\mathcal{N}) - (-g - 1 + h^0(\mathcal{N})) = \deg \mathcal{N} - g + 1$$

$$\implies \text{oops! I lost } h^0(\mathcal{N}) \text{ in this operation...}$$

Maybe if I just use normal exact sequence and realise that

$$h^0(\mathcal{N}) = h^0(\mathcal{T}_X|_C) - h^0(\mathcal{T}_C)$$

I know that $h^0(\mathcal{T}_C) = 0$ for $g > 1$, so the question is how to compute the restricted holomorphic vector fields. □

6. Deformations of Stanley-Reisner schemes

My aim in this section is to cover the minimal theory needed to apply results in [**?**] to my deformation problem. The main theorem I expect to use is the following:

Theorem 6.1. *If $\mathcal{K}$ is a 3-dimensional manifold then*

$$\dim T^1_{\mathbb{P}(K)} = 11d_3 + 5e_3 + 3e_4 + e_{\geq 5} + c_{\geq 6} + 5f_1^{(3)} + 2f_1^{(4)} + h^2(\mathcal{K}).$$

I need to do some interpretation first. From [**?**, p. 1]:

> "For each $\mathbb{C}$-algebra A, there is a cohomology theory providing modules T_A^i. However, only T_A^1 and T_A^2 are relevant for the deformation theory of $\operatorname{Spec} A$. The module T_A^1 collects the infinitesimal deformations of A and T_A^2 contains the obstructions for lifting these deformations to decent parameter spaces. Eventually, $\operatorname{Der}_{\mathbb{C}}(A, A)$ will be called T_A^0."

[**?**, p. 3]:

> "In general, for a finitely generated $\mathbb{C}$-algebra A, the modules T_A^i allow the following ad hoc definitions: Let $P = \mathbb{C}[x_0, \ldots, x_n]$ mapping onto A so that $A \simeq P/I$ for an ideal I. The T_A^1 is the cokernel of the natural map $\operatorname{Der}_{\mathbb{C}}(P, P) \to \operatorname{Hom}_P(I, A)$. Moreovr, [...] and we obtain T_A^2 as the cokernel of the induced map $\operatorname{Hom}_P(P^m, A) \to \operatorname{Hom}_A(R/R_0, A)$."

Remark 6.2 (dani). Let's think more about that definition of T_A^1 as the cokernel of the "natural map" $\operatorname{Der}_{\mathbb{C}}(P, P) \to \operatorname{Hom}_P(I, A)$.

Pick a derivation $\chi \in \operatorname{Der}_{\mathbb{C}}(P, P)$. We want a map $I \to A = P/I$. Pick an element $a \in I$ and map it to $\chi(a)$. And that's it. It might be zero or not. If the χ we picked maps any $a \in I$ to some element in I, then this χ is in the kernel of the natural map. So T_A^1 is the image of the natural map modulo the kernel.

Unfortunately this doesn't say much about the following crucial fact: the T_A^i modules are $\mathbb{Z}^{n+1}$-graded. But here goes an explanation. We use the notation that e_p is a "face not in the complex", which of course corresponds to a generating monomial x^p of the SR ideal I. It has a degree, of course (given by the "support" of p). Now here's a rather sloppy step but: since the cotangent modules are built upon I, who is given by e_p, we just notice that the degree of e_p along with a (very natural, too) notion of degree among relations, induce a grading in all the cotangent modules T_A^i. Thus, the grading is morally just the degree of the monomials, which is in fact the support of p, a vector $\mathbf{c} \in \mathbb{Z}^{n+1}$.

Next in line is the decomposition $\mathbf{c} = \mathbf{a} - \mathbf{b}$ into positive minus negative part. But how exactly...?

7. Miscellaneous

Look for the "Perfect obstruction theory". This is used by Bruzzo for deformations of (super)stacks, work with E. Paiva and —.